MLP: I6 and I5 versus unchanged Adam

First-linear weights + biases frozen at 448; learning rate changes at 500; same 5,000 examples.

Baseline: unchanged Adam I6: freeze + LR 0.0025 I5: freeze + LR 0.005

I6: freeze + LR 0.0025 from 500

I5: freeze + LR 0.005 from 500

(a) Full training loss

0.072448 at 707

2.5858 at 505

(b) $\widehat{\eta\kappa}_t$ (symlog)

(c) Gradient cosine

(d) Full $\|g_t\|$ (log)

(e) Later-layer $\|g_t\|$ (log)

(f) Output-change cosine

(g) Output RMS (log)

(h) LR of later layers

Completed optimizer updates t

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Training loss: independent linear ranges. Other rows share scales; dotted line: LR change at 500.
Row (c): solid = full raw gradient (656,810 parameters); dashed = later layers (42,210), also used in row (e).
$\widehat{\eta\kappa}_t = 1 - \langle g_t, g_{t-1}\rangle / \|g_{t-1}\|^2$ is a projected-gradient diagnostic under Adam. Output changes are outgoing.